

EXPERIMENT NO. - 3

Object :- (A) To prepare the $\bar{X}$ and $\bar{R}$ chart for the given sample of product

(B) To show that distribution of sample mean from any universe follow normal distribution. Further analyse the effect of sample size on distribution

Theory :-

A Control Chart can be defined as a chronological graphical comparison of actual product quality characteristics with the information gathered during inspection of products

Types of Control Chart :-

- (i) $\bar{X}$ and $\bar{R}$ Chart
- (ii) P chart (based on fraction defectives)

$\bar{X}$ and $\bar{R}$ Chart - These charts are known as Control Chart for variables. Since these charts give the control limit for measuring the variation in quality characteristics. The basic principle of measurement of control chart limit is similar to that of computing

the frequency distribution process

Procedure :-

- Select the appropriate quality characteristics to be studied
- Record data : required no of samples compared to an adequate no. of parts and size
- Determine the control limit from the data
- Check if these control limits are economically satisfactory for the job.
- Plot the control limit on suitable graph paper
- Start the plotting of recorded result of production sample within or exceeding the control limit.

Formulae used :-

• For $\bar{X}$ Chart :

When range is used as a measure of spread,

- Centre line - $\bar{\bar{X}} = \Sigma \bar{X} / n$
- U.C.L. (upper control limit) - $\bar{\bar{X}} + A_2 \bar{R}$
- L.C.L. (lower) - $\bar{\bar{X}} - A_2 \bar{R}$

where,

n = total sample

A_2 = constant

$$\bar{R} = \Sigma R / n$$

Calculations :-

→ no. of samples = 5 (sample size)

→ $n = 20$

→ $A_2 = 0.58$

Factor for $\bar{R}$ chart, $D_3 = 0$, $D_4 = 2.11$

→ Control line,

$$\bar{\bar{X}} = \frac{185.94}{20} = 9.297 \approx 9.3.$$

$$\bar{R} = \frac{\sum R}{n} = \frac{2.17}{20} = 0.108 \approx 0.11$$

→ For $\bar{X}$ Chart,

$$\begin{aligned} \text{U.C.L.} = \bar{\bar{X}} &= \bar{\bar{X}} + A_2 \bar{R} \\ &= 9.3 + 0.58 \times 0.11 \\ &= 9.363 \end{aligned}$$

$$\begin{aligned} \text{L.C.L.} = \bar{\bar{X}} &= \bar{\bar{X}} - A_2 \bar{R} \\ &= 9.3 - 0.58 \times 0.11 \\ &= 9.236 \end{aligned}$$

→ For $\bar{R}$ Chart,

Centre line, $\bar{R} = 0.11$

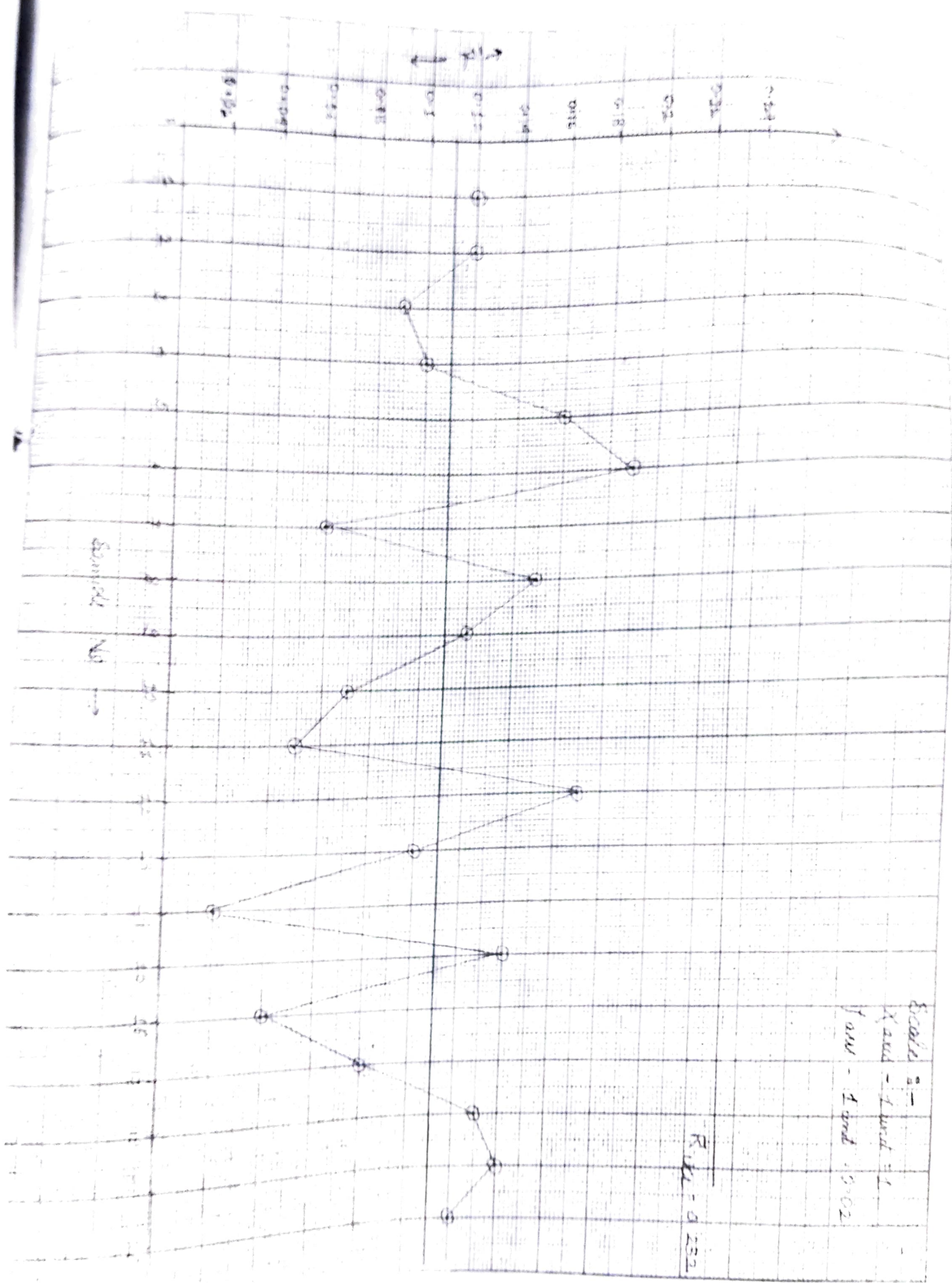
$$\text{U.C.L.}_R = D_4 \bar{R} = 2.11 \times 0.11 = 0.232$$

$$\text{L.C.L.}_R = D_3 \bar{R} = 0 \times 0.11 = 0$$

Scale 1-
 X axis - 1 unit = 1
 Y axis - 1 unit = 0.02

R.H. = 0.252

Sample No. →

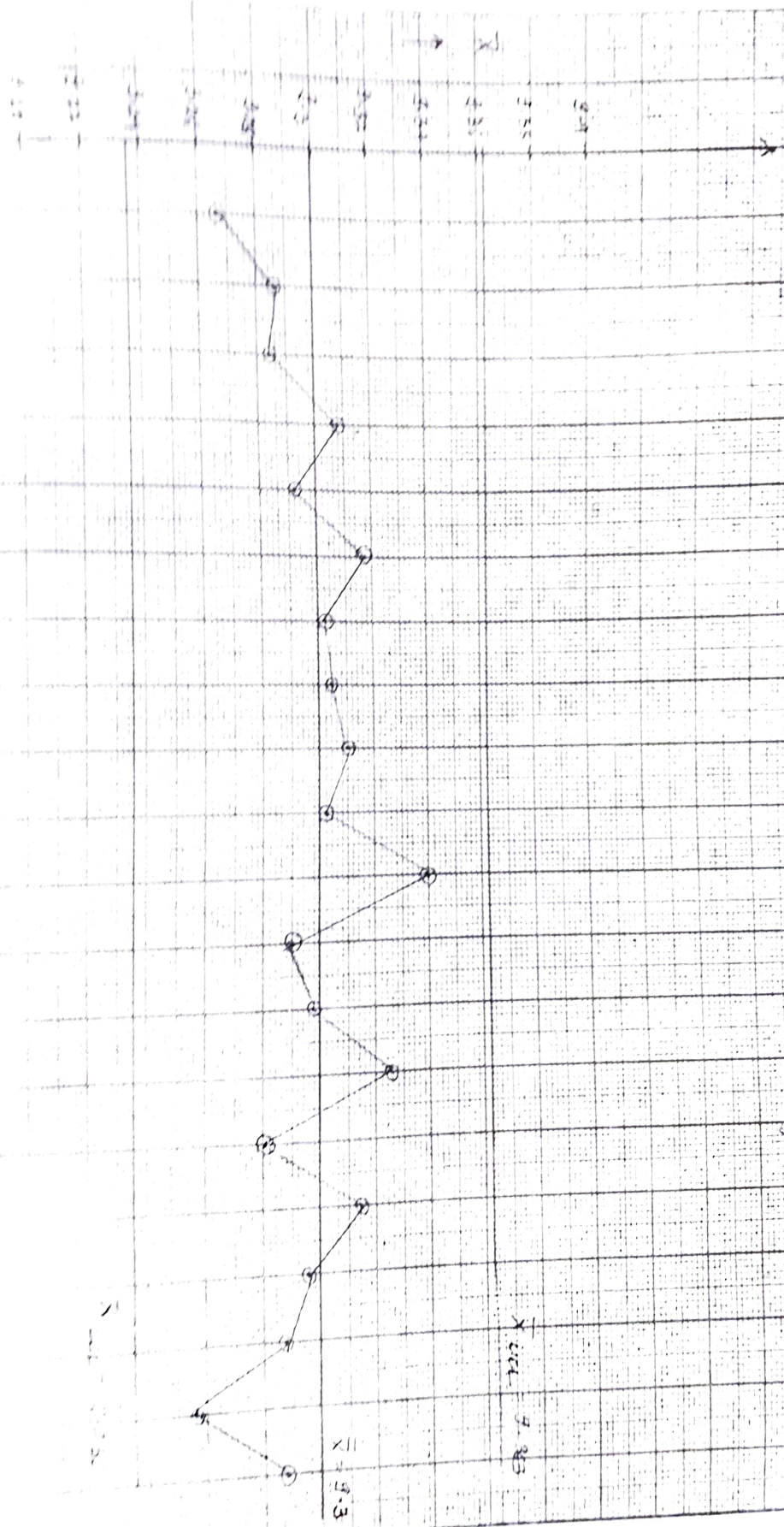


X (Good for (N=20))

Sale
 1 unit = 1 pump
 1 unit = 1 unit
 1 unit = 1 unit

X 100 5 25

X 100 5 3



→ For $\bar{R}$ Chart :

(i) Centre line - $\bar{R} = \Sigma R / K$

(ii) U.C.L. (R) - $D_4 \cdot \bar{R}$

(iii) L.C.L. (R) - $D_3 \cdot \bar{R}$

(K = sub group)

where,

D_3 and D_4 are constants that depends upon the sample size from which that particular sample is obtained.

Observations :-

A) Sample Size = 5

S.No	Sample Elements					Average $\bar{X} = \Sigma x / n$	Range ($X_{max} - X_{min}$)
	X_1	X_2	X_3	X_4	X_5		
1	9.18	9.28	9.28	9.29	9.3	9.266	0.12
2	9.29	9.32	9.31	9.2	9.31	9.286	0.12
3	9.32	9.31	9.24	9.32	9.23	9.284	0.09
4	9.25	9.32	9.35	9.31	9.31	9.308	0.1
5	9.33	9.33	9.31	9.32	9.17	9.292	0.16
6	9.37	9.38	9.33	9.19	9.32	9.318	0.19
7	9.27	9.32	9.27	9.32	9.33	9.302	0.06
8	9.33	9.19	9.34	9.34	9.32	9.304	0.15
9	9.35	9.33	9.23	9.32	9.32	9.31	0.12
10	9.33	9.26	9.28	9.32	9.32	9.302	0.07
11	9.31	9.35	9.36	9.36	9.32	9.34	0.05

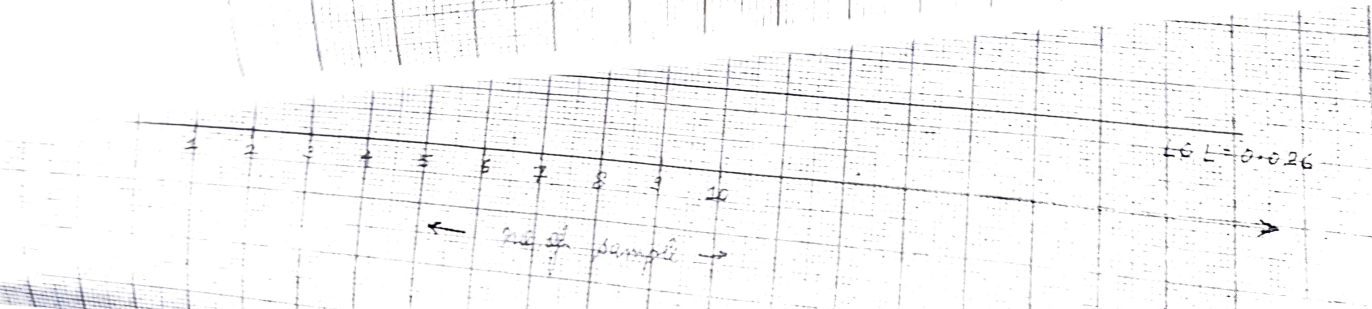
S.Ns.	Sample elements					Average $\bar{X} = \Sigma x/n$	Range ($X_{\max} - X_{\min}$)
	X_1	X_2	X_3	X_4	X_5		
12	9.17	9.26	9.34	9.33	9.34	9.288	0.17
13	9.28	9.33	9.23	9.32	9.33	9.298	0.1
14	9.32	9.34	9.33	9.32	9.32	9.326	0.02
15	9.18	9.27	9.32	9.31	9.32	9.28	0.14
16	9.31	9.32	9.33	9.32	9.29	9.314	0.04
17	9.25	9.33	9.33	9.32	9.25	9.296	0.08
18	9.34	9.23	9.33	9.34	9.2	9.288	0.13
19	9.31	9.23	9.33	9.19	9.22	9.256	0.14
20	9.26	9.31	9.21	9.33	9.33	9.288	0.12
						$\Sigma \bar{X} = 185.94$	$\Sigma R = 2.17$

(B) Sample Size - 10

S.No.	Sample elements										Average $\bar{X} = \Sigma x/n$	Range $X_{\max} - X_{\min}$
	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}		
1)	9.24	9.33	9.33	9.32	9.33	9.25	9.32	9.32	9.33	9.33	9.338	0.09
2)	9.34	9.2	9.33	9.22	9.24	9.23	9.33	9.33	9.33	9.32	9.28	0.14
3)	9.32	9.27	9.19	9.23	9.2	9.33	9.34	9.18	9.32	9.33	9.26	0.16
4)	9.33	9.32	9.32	9.34	9.26	9.33	9.33	9.31	9.24	9.25	9.303	0.1
5)	9.32	9.35	9.34	9.24	9.34	9.37	9.23	9.23	9.33	9.32	9.307	0.12



9.16 9.18 9.2 9.22 9.24 9.26 9.28 9.3 9.32
 9.34 9.36 9.38 9.4 9.42 9.44 9.46 9.48 9.5 9.52
 9.54 9.56 9.58 9.6 9.62 9.64 9.66 9.68 9.7 9.72
 9.74 9.76 9.78 9.8 9.82 9.84 9.86 9.88 9.9 9.92
 9.94 9.96 9.98 10.0



$LCI = 0.026$

high sample

Expt 5-

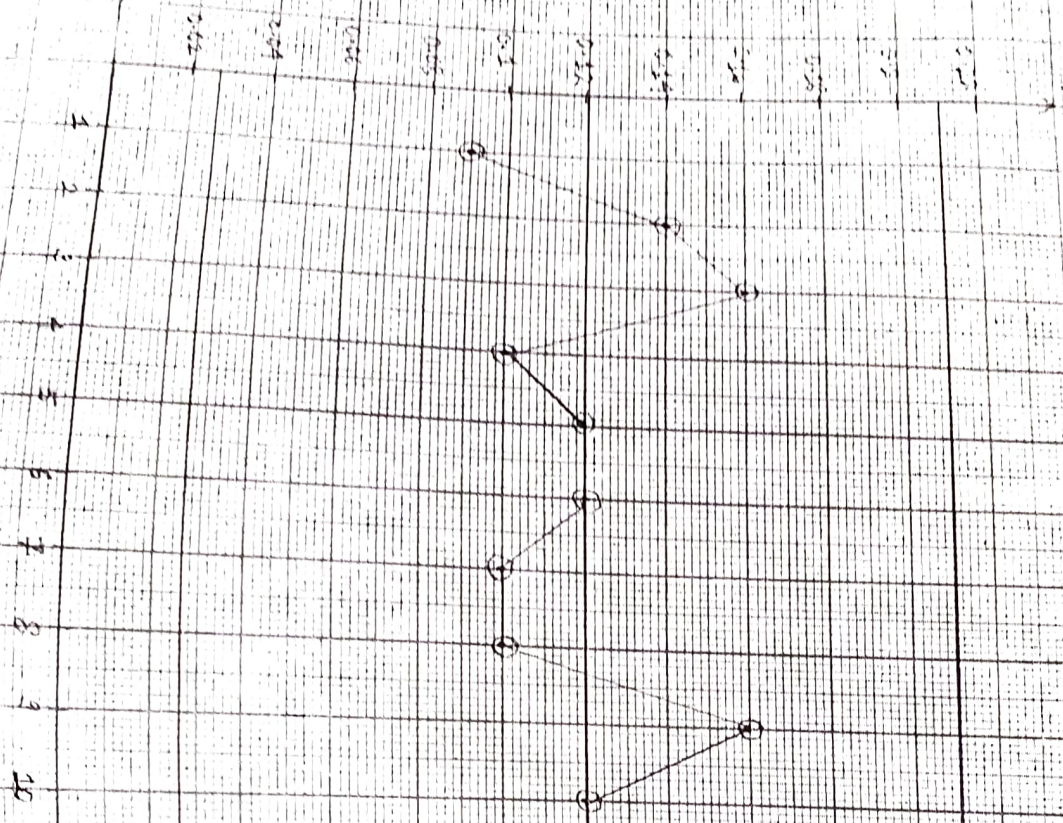
$X_{avg} = 1$

$y_{avg} = 0.02$

$$W_L = 0.24$$

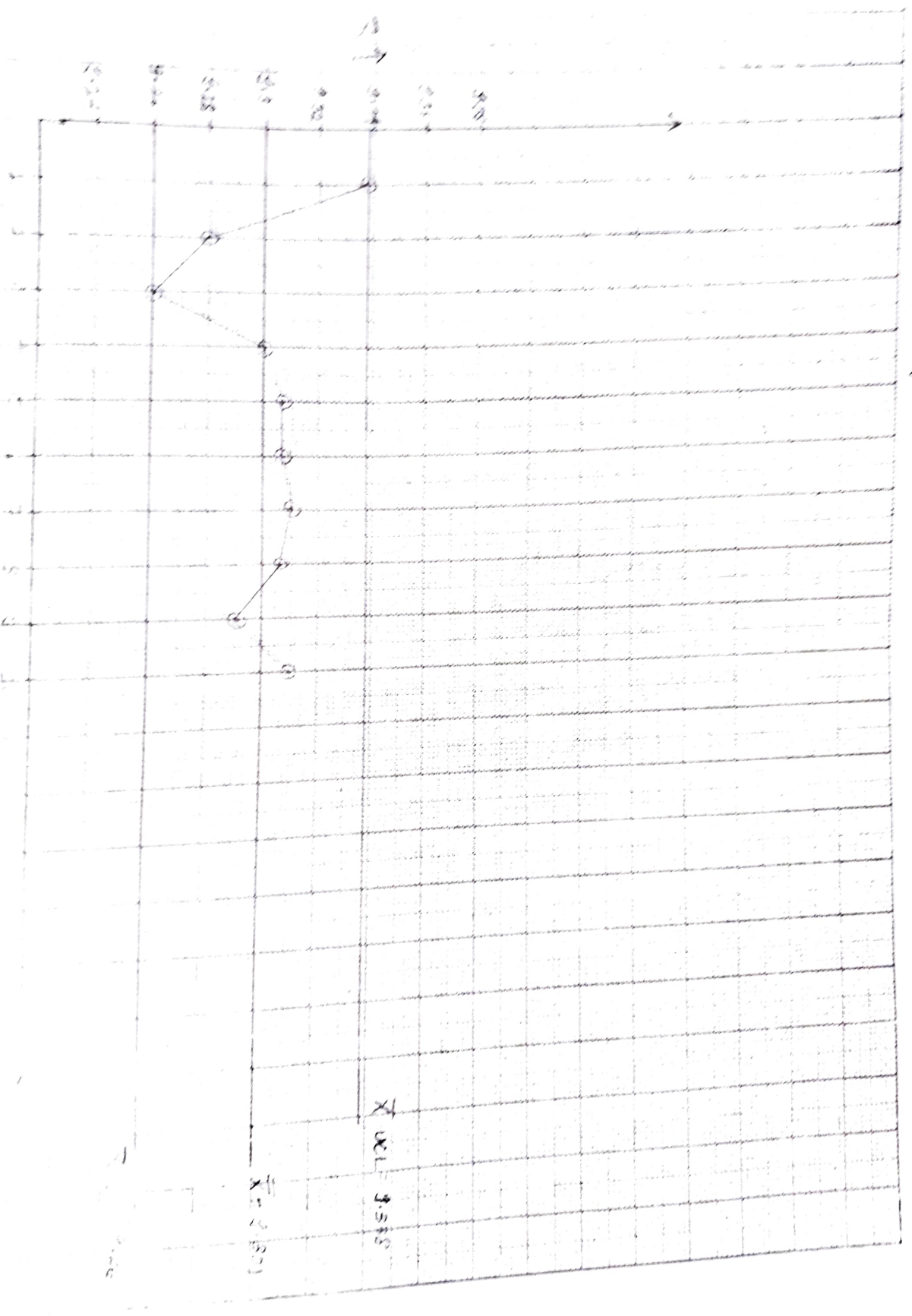
$$\bar{R} = 0.121$$

$$L_6 L = 0.026$$



← No. of sample →

Hand for 1.5.1



X = 3.5

X = 3.0

S.No.	Sample Elements										Average ($\bar{X}$)	Range (R)
	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}		
6)	9.33	9.33	9.24	9.23	9.32	9.33	9.33	9.34	9.27	9.35	9.307	0.12
7)	9.34	9.32	9.32	9.32	9.33	9.33	9.26	9.24	9.32	9.32	9.31	0.1
8)	9.32	9.34	9.32	9.32	9.24	9.33	9.32	9.32	9.24	9.32	9.307	0.1
9)	9.19	9.17	9.33	9.33	9.26	9.32	9.33	9.33	9.33	9.32	9.29	0.16
10)	9.26	9.35	9.33	9.31	9.34	9.28	9.34	9.33	9.23	9.34	9.311	0.12
											$\bar{\bar{X}} = 9.301$	$\bar{R} = 0.121$

→ $\bar{X}$ Chart, $n=10$, $\bar{\bar{X}} = 9.301$

$$A_2 = 0.31$$

$$\begin{aligned} \text{U.C.L.} &= \bar{\bar{X}} + A_2 \bar{R} = 9.301 + 0.31 \times 0.121 \\ &= 9.338 \end{aligned}$$

$$\begin{aligned} \text{L.C.L.} &= \bar{\bar{X}} - A_2 \bar{R} = 9.301 - 0.31 \times 0.121 \\ &= 9.26 \end{aligned}$$

→ R Chart, $n=10$, $D_4 = 1.78$, $\bar{R} = 0.121$

$$\begin{aligned} \text{U.C.L.} &= D_4 \times \bar{R} = 1.78 \times 0.121 \\ &= 0.21 \end{aligned}$$

$$\begin{aligned} \text{L.C.L.} &= D_3 \times \bar{R} = 0.22 \times 0.121 \\ &= 0.0266 \end{aligned}$$

Normal Distribution of Sample Size :-

Theory :-

The Normal Distribution Curve is indicated and drawn using histograms. It is plotted with the help of frequency of the sample mean in the given interval of value. The mid point of histograms combined together and this results in a curve which follows a normal distribution curve.

Procedure :-

- 1) Select the quantity to be measured.
- 2) Take suitable amount of indicated piece to take measurement.
- 3) Grouping them in sample having a suitable amount of sample size.
- 4) Find out the mean of samples.
- 5) By collecting different sample mean, choose the interval.
- 6) Find out the frequency of the individual interval in between the value of sample mean exist.
- 7) Draw the histogram.
- 8) Joint the mid point of the histogram.
- 9) Analyze the curve.

NORMAL DISTRIBUTION CURVE

Scale :-

X axis - 1 unit = 0.02

Y axis - 1 unit = 4

